```
# Step 1: Secure System Dependency Allocation
!pip install -q ultralytics roboflow opencv-python numpy matplotlib

```

```
# Step 2: Framework Importing & Hardware Optimization Check
import os
import cv2
import torch
import numpy as np
import matplotlib.pyplot as plt
from ultralytics import YOLO
from roboflow import Roboflow

device = "cuda" if torch.cuda.is_available() else "cpu"
print(f"[System Initialization] Target compute engine mapped to: {device.upper()}")

model = YOLO('yolo11n.pt')
print("[Framework Status] YOLOv11 nano baseline structural layers successfully initialized.")

```

```
[System Initialization] Target compute engine mapped to: CUDA
[Framework Status] YOLOv11 nano baseline structural layers successfully initialized.
```

```
# Step 3: Programmatic Real-World Dataset Extraction
# Authenticated direct cloud streaming from a verified open universe repository
ROBOFLOW_API_KEY = "ciGGavgvfgz1x8UOAPx7"
WORKSPACE_NAME = "ntut-fkont"
PROJECT_NAME = "dirty-detect"
VERSION_NUMBER = 1

try:
    rf = Roboflow(api_key=ROBOFLOW_API_KEY)
    project = rf.workspace(WORKSPACE_NAME).project(PROJECT_NAME)
    version = project.version(VERSION_NUMBER)

    # Download and extract the real-world dataset directly into the sandbox instance
    dataset = version.download("yolov11")
    data_yaml_path = os.path.join(dataset.location, "data.yaml")
    print(f"[Data Setup] Success! Real-world dataset extracted to cloud storage: {dataset.loc
except Exception as e:
    print(f"[Data Warning] Configuration error, falling back to basic schema. Details: {e}")
    data_yaml_path = "coco8.yaml"

```

```
loading Roboflow workspace...
loading Roboflow project...
[Data Setup] Success! Real-world dataset extracted to cloud storage: /content/Dirty-Detect-1
```

```
# Step 4: Full Multi-Epoch Production Training Loop
print("[Training Started] Running 50-epoch backpropagation sequence...")
training_results = model.train(
    data=data_yaml_path,
    epochs=50,
    imgsz=640,
    device=device,
    project="SentinelCleanAIFN_Build",
    name="production_run",
```

```
    verbose=True
)
print("[Training Complete] Custom weights generated. Checkpoint matrices compiled.")

```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      43/50       2.7G      1.135      1.277      1.701          4        640: 100% ━━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━
                   all         69         74      0.365      0.349      0.334      0.191

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      44/50       2.7G      1.119      1.208      1.711          6        640: 100% ━━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━
                   all         69         74      0.259      0.269      0.327      0.202

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      45/50       2.7G       1.08      1.135       1.63          6        640: 100% ━━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━
                   all         69         74      0.201      0.393      0.285      0.157

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      46/50       2.7G      1.084      1.155      1.636          4        640: 100% ━━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━
                   all         69         74      0.265      0.275      0.303      0.179

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      47/50       2.7G      1.051      1.128       1.62          4        640: 100% ━━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━
                   all         69         74      0.305      0.259      0.307      0.178

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      48/50       2.7G      1.027      1.061      1.596          4        640: 100% ━━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━
                   all         69         74      0.315      0.302       0.26       0.14

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      49/50       2.7G      1.002      1.049      1.564          4        640: 100% ━━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━
                   all         69         74      0.325      0.364      0.304      0.175

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      50/50       2.7G     0.9924      1.062      1.573          4        640: 100% ━━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━
                   all         69         74      0.317      0.316      0.343      0.197

50 epochs completed in 0.229 hours.
Optimizer stripped from /content/runs/detect/SentinelCleanAIFN_Build/production_run-2/weights/la
Optimizer stripped from /content/runs/detect/SentinelCleanAIFN_Build/production_run-2/weights/be

Validating /content/runs/detect/SentinelCleanAIFN_Build/production_run-2/weights/best.pt...
Ultralytics 8.4.115 🚀 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO11n summary (fused): 101 layers, 2,583,322 parameters, 0 gradients, 6.4 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━
                   all         69         74      0.254      0.269      0.327      0.201
          Brown Stains          4          4      0.444       0.75       0.77      0.598
                 Other          2          4          0          0     0.0273    0.00637
            Red Stains          4          4      0.391       0.25       0.34      0.193
            Tea Stains          2          3          0          0     0.0511     0.0141
          Water Stains         56         57      0.691      0.614      0.678      0.329
          White Stains          2          2          0          0     0.0951     0.0666
Speed: 0.2ms preprocess, 5.4ms inference, 0.0ms loss, 4.2ms postprocess per image
Results saved to /content/runs/detect/SentinelCleanAIFN_Build/production_run-2
[Training Complete] Custom weights generated. Checkpoint matrices compiled.
```

```
# Step 5: Metrics Extraction & Automated Results Export
print("\n--- Final System Analysis Matrix (Honest Performance Reporting) ---")
```

```
evaluation_metrics = model.val(device=device)

print(f"📊 Achieved Mean Average Precision [mAP@50-95]: {evaluation_metrics.box.map:.4f}")
print(f"📊 Achieved Mean Average Precision [mAP@50]: {evaluation_metrics.box.map50:.4f}")
print(f"🎯 Comprehensive System Recall Margin: {evaluation_metrics.box.mr.mean():.4f}")

# Create the results directory and systematically save validation charts
os.makedirs("../results", exist_ok=True)
!cp -r SentinelCleanAIFN_Build/production_run/* ../results/ 2>/dev/null
print("[Storage Output] Authentic model training graphs and matrices copied to results/ folder
```

```
--- Final System Analysis Matrix (Honest Performance Reporting) ---
Ultralytics 8.4.115 🚀 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLO11n summary (fused): 101 layers, 2,583,322 parameters, 0 gradients, 6.4 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1415.3±561.0 MB/s, size: 35.4 KB)
val: Scanning /content/Dirty-Detect-1/valid/labels.cache... 69 images, 0 backgrounds, 0 corrupt: 1
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━
                   all         69         74      0.252       0.27      0.327      0.201
           Brown Stains          4          4       0.43      0.754       0.77      0.598
                 Other          2          4          0          0     0.0273    0.00637
             Red Stains          4          4      0.391       0.25       0.34      0.193
             Tea Stains          2          3          0          0     0.0511     0.0169
           Water Stains         56         57      0.692      0.614      0.678      0.328
           White Stains          2          2          0          0     0.0945     0.0662
Speed: 13.2ms preprocess, 7.3ms inference, 0.0ms loss, 4.3ms postprocess per image
Results saved to /content/runs/detect/val-2
📊 Achieved Mean Average Precision [mAP@50-95]: 0.2014
📊 Achieved Mean Average Precision [mAP@50]: 0.3268
🎯 Comprehensive System Recall Margin: 0.2698
[Storage Output] Authentic model training graphs and matrices copied to results/ folder.
```

```
# --- STEP 6: PRODUCTION LIVE INFERENCE DEMO VIDEO
from ultralytics import YOLO
import matplotlib.pyplot as plt
import os
import cv2

# Load the custom fine-tuned weights
trained_model = YOLO('/content/runs/detect/SentinelCleanAIFN_Build/production_run/weights/best

# Target the pre-separated real validation images folder downloaded in Step 3
test_image_dir = '/content/Dirty-Detect-1/valid/images'
sample_images = [os.path.join(test_image_dir, f) for f in os.listdir(test_image_dir) if f.ends

if len(sample_images) > 0:
    # Select the first real floor stain image from the folder
    demo_target_path = sample_images[0]
    print(f"[Demo Core] Processing real floor input imagery: {os.path.basename(demo_target_pat

    # Run instantaneous onboard edge inference
    demo_results = trained_model(demo_target_path, conf=0.25, verbose=False)

    # Render and display the real vision output box on screen
    for result in demo_results:
        annotated_img = result.plot()
        annotated_img_rgb = cv2.cvtColor(annotated_img, cv2.COLOR_BGR2RGB)

        plt.figure(figsize=(8, 8))
        plt.imshow(annotated_img_rgb)
        plt.axis('off')
```

```
        plt.title(f"SentinelCleanAIFN Active Edge Output: {result.boxes[0].cls.tolist() if ler
        plt.show()
else:
    print("[Demo Error] Run Step 3 first to extract the test split folder.")

```

```
[Demo Core] Processing real floor input imagery: images-30-_jpg.rf.03c0f268b61a7a269cae8077263e946
```